

“Dance of Drums”: An Interactive Installation of ICH Dance Representation Through the Combination of Virtual and Reality

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Figure 1: (a) The participant beats an interactive target of the projected interface (b) The triggered virtual feedback in the initial VR scene.

ABSTRACT

With the development of the digital information, the digitalization and creative representation of the intangible culture heritage (ICH) dances have become the core of dance safeguarding. “Dance of Drums”, an interactive installation combining virtual and reality for Chinese traditional bronze-drum dance representation, provides both external projected interface and internal virtual reality (VR) experiences. Specifically, “Dance of Drums” rebuilds the motions and folklore elements of bronze-drum dance through internal VR scenarios, and shows the history and ancient form of dance through the projected interface in real space, which reconstructs the static display of dances to promote the preservation and dissemination of bronze-drum dance.

CCS CONCEPTS

• Applied computing; • Arts and humanities; • Performing arts; • Human-centered computing; • Interaction design; • Interaction design process and methods;

KEYWORDS

Intangible cultural heritage, dance representation, VR installation, interaction design

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1 INTRODUCTION

The bronze-drum dance is an ancient circle dance inherited among the Zhuang and Yi people in Yunnan Province, China. As the primary environment changes, the ICH dance cannot be directly understood, the inheritance and safeguarding facing challenges [Ma et al., 2019]. Nowadays, the digital technology for safeguarding ICH dances mainly applies in four domains: choreography, motion capture, digital education, and creative representation and dissemination. For instance, Takeshi Miura et al. estimated a traditional Bon Odori dance style lost in the Japan [Miura et al., 2015]. Moreover, creative digitalization works deeply promote the ICH dances safeguarding. For example, focusing on *Mazu* culture in China, Jia Ming day et al. explored a virtual interactive experience “*White Crane Dance*” by reconstructing the festival scene of this folk dance [Day et al., 2018]. It could be seen that there was a rising trend of carrying out VR in the ICH dance domain. For the representation and dissemination of ICH dances, VR application has become a new approach to digitally safeguard and creatively display [Lu et al., 2021].

2 DESIGN AND IMPLEMENTATION

“Dance of Drums” designed an arc-shaped space, where could accommodate up to two participants, who interacted with either the VR scenarios or the projected interface. For the projected interface, we produced dancing animation clips for drumming interactions, which the participants could use physical drumstick to beat the targets that projected on a curtain. The sensing of beating and animations activation were done by the vibration sensors installed

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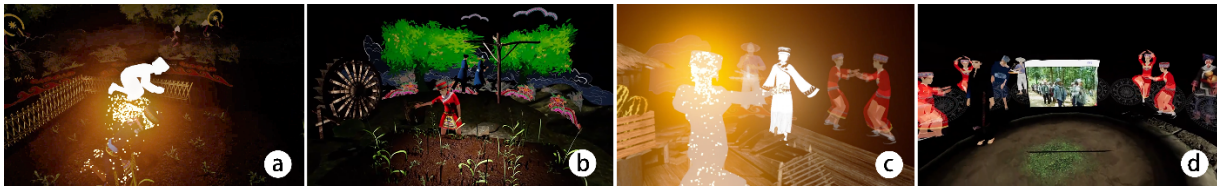


Figure 2: (a) Spring dance scenario: sowing motion (b) Summer dance scenario: *Moqiu* motion (c) Autumn dance scenario: harvesting motion and group dance (d) Winter dance scenario: dancing square appears the popularization video

on the curtain. Meanwhile, the corresponding virtual feedback inside the VR scenarios might change when the external projected targets were activated. The trigger device was implemented by Vive Tracker installed on the drumstick's head side for real-time detecting the location of the external participant. In VR, we used HTC Vive and Unreal Engine 4 to create virtual dancing scenarios in which the participants could enter to view dance motions in sowing, *MoQiu*, harvesting and *ChongMi* laboring.

2.1 The Combination of Virtual and Reality

Asymmetric interactive experiences of multi-VR users' collaboration such as [Gugenheimer et al., 2017][Welsford-Ackroyd et al., 2021] inspired us to combine virtual and reality. We designed to link interactions between VR and non-VR participants, which made the movements of external non-VR participants to be parts of the VR scenes. In the initial VR scene, the participants are placed on top of a giant bronze-drum with specific patterns of bronze-drum around. When the external participant picks up the physical drumstick to tap the projected targets and move to different drumming locations, specific patterns in VR around the bronze-drum are triggered and lighted up (See Figure 1). When the VR participant moves close to the red animated patterns in time, some popularizing information of drum patterns and the folklore elements of bronze-drum dance would show up.

2.2 VR Scenarios

The entire VR scenarios focus on symbolic dance motions and dancing space. In the initial scene, the participant can pick up the magnifying glass, and select four narrative patterns in the center of the scene to enter "*Dance of the Four Seasons*" scenarios (See Figure 2). In the spring dance scenario, a virtual female character will lead the participant to a field to follow the hint stroking the farm cattle, and then give a seed bag for sowing the seeds to the field. When the interaction completes, *Yi* will perform the action of sowing and burying the soil, then her action will gradually fade and just retain the particle effect of the motion trajectory. The hand flinging action related to *Yi*, *Zhuang* labor habits in the bronze-drum dance will be displayed, after that the dancing motion explanation will appear. In the summer dance scenario, the virtual character is spreading water and leading the participant to a position of the swing, where participant can view the folklore activity of "*Moqiu*" (*Yi* summer custom of swinging). In the autumn dance scenario, the virtual character narrates harvest story to lead participant to help her carry baskets and bronze-drum, then the virtual character activated by participant will dance together to show the joy of harvesting. The winter dance scenario reproduces a traditional Chinese New

Year situation, and the character sweeps the floor. The participant picks up a wooden handle to "*ChongMi*" (Grinding rice), after that the virtual character will point in the direction of *Gongping* dancing square, the participant moves to the dancing square and listens to the bronze-drum dance introduction. At the end of the experience, the VR participant is surrounded by the dancing crowds, a video of bronze-drum dance popularization is showed to strengthen the memory.

3 CONCLUSION

This paper summarizes the methods of representing bronze-drum dance in VR and designs interactive experience that combines virtual and reality. We reproduce the bronze-drum dance situation by focusing on the dance motions and dancing space that related to local labor activities. The design of the interactive installation provides a stronger sense of presence and ritual for the dance representation and gives a primary sample for future ICH dance dissemination in VR.

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